LUCAS GONÇALVES

AI Engineer | LLM Agents · RAG · Data Pipelines on GCP

Brazil (GMT-3) · Remote-ready, any time zone · Open to relocation

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SUMMARY

AI Engineer with 2+ years building retrieval-augmented and agentic LLM systems in Python with FastAPI, LangGraph, Qdrant and pgvector. Built a self-hostable RAG service that ships a semantic-entropy confidence score with every answer, calibrating its agent domain gate to 96.7% true-positive at 3.3% false-positive over a 120-question labeled sweep. Drove the retrieval and grounding core of an AI advising platform to 1,149 automated tests, and re-architected a multi-agent code-generation platform behind a deterministic security gate. Targeting AI Engineer roles in RAG, agent orchestration and applied LLM systems.

KEY ACHIEVEMENTS

- **Calibrated an LLM agent's domain gate from measurement rather than intuition**: an ROC sweep over 120 labeled questions set it at 96.7% true-positive / 3.3% false-positive, frozen as a fixture so CI fails if a default drifts.
- **Co-led a two-person effort that cut an incorrect field-parts return rate from ~20% in 2025 to 0% by June 2026**, against a 6% Six Sigma target, removing recurring freight cost from unnecessary returns across a 150+ dealership network.
- **Improved sentiment classification to 0.887 macro F1 against a 0.584 frozen-features baseline**, a 51.9% relative gain concentrated in the rarest classes, driven by per-class error analysis.

EXPERIENCE

Unipac | Software Engineer - AI & Data | 09/2026 - Present | Pompeia, SP, Brazil (Hybrid)

Packaging and polymer processing manufacturer within the Jacto Group. Internal AI platform for automated software generation.

- Re-architected a multi-agent code-generation platform, replacing an LLM orchestrator with a deterministic 4-wave scheduler for up to 12 specialist agent phases. *(LangGraph, LangChain, FastAPI, GCP Cloud Run)*
- Piloted it on 22 InovaSkill 2026 projects: 21 of 21 severe findings blocked before delivery and 12 of 14 end-to-end runs passed.
- Replaced the model's self-reported security verdict with a deterministic gate: Semgrep (20 custom rules) and Trivy in a read-only sandbox, judged by a pure policy function that blocks critical and high findings.
- Engineered a 9-layer LLM agent guardrail stack (caps, loop breaker, retry, fallback, circuit breaker) after one model response emitted 2,179 identical tool calls.

PM Accelerator | AI Engineer | 06/2026 - 08/2026 | Miami, FL (Remote)

Ed-tech startup within PM Accelerator. AI college-advising platform (Collegiary).

- Built a hybrid retrieval engine from scratch, with no RAG framework, over ~1,800 US institutions: exact name and alias matches rank above pgvector similarity by design, and an explicit insufficient-data signal replaces the answer below the relevance threshold.
- Eliminated a class of LLM hallucination structurally rather than by prompt wording: a gate that refuses before the API call, spending neither tokens nor quota, and a verifier that flags figures no retrieved record supports, including one institution's statistic quoted under another's name.
- Verified prompt rules and the web-search fallback against a live model instead of fixtures: the source-declaration line held on 7 of 7 turns across six question kinds, with unreached paths documented as unproven.
- Scaled the backend suite to 1,149 automated tests at a 1.75:1 test-to-code ratio, and cut the CI dependency footprint ~9x (~800MB to ~90MB) by serving the embedding model through ONNX instead of a deep-learning framework.
- Delivered 19 of 29 tracked board items on a weekly cadence against written specifications, and was recommended in writing by the company's founder, Dr. Nancy Li, for technical skills and ownership.

Jacto | After-Sales - Data & Automation | 11/2024 - 06/2026 | Pompeia, SP, Brazil (Hybrid)

Global agricultural machinery manufacturer, 5,000+ employees, operating in 100+ countries.

- Automated fiscal-address collection from a foreign government registry in Python (Selenium, pandas): one run covers 99 regional pages, translates them and exports a load-ready workbook for Salesforce, removing up to 5 hours a day of manual work (~25 to 100 hours per collection).
- Co-led a two-person effort that cut the incorrect field-parts return rate from ~20% in 2025 to 0% by June 2026, with an approved manual adopted network-wide and an estimated R$100-135K in freight avoided in 1H 2026.
- Tracked and validated 20+ improvements with the developers of an internal parts-return system, including a redesign that cut record loading from 5-10 minutes to under 1 minute.

CIAg | Software Engineer - Synapse Program | 02/2024 - 10/2024 | Pompeia, SP, Brazil (Hybrid)

Agribusiness innovation center within the Jacto Group. Synapse program: LLM assistant for a domain-specific knowledge base.

- Led a team of 2-3 developers building an LLM assistant for students and faculty (Python, FastAPI, OpenAI API), answering from agronomic documents through contextual RAG and a REST layer between the application and the model.

PROJECTS

SmartB100: Self-hostable RAG API with semantic-entropy hallucination scoring

Python · FastAPI · Qdrant · Ollama · LangGraph · Docker · github.com/LukeSantossz/sb100_agents

- Architected a self-hostable RAG service on FastAPI, Qdrant and Ollama, with every significant trade-off recorded in 15 Architecture Decision Records that name and reject the alternatives considered.
- Implemented semantic-entropy hallucination scoring (Farquhar et al., 2023): N-sample generation, embedding-similarity clustering with a cache reducing the pass from $O(N^2)$ to N embed calls, and normalized Shannon entropy, behind a gate that degrades to a neutral score rather than failing.
- Reached 372 automated tests at 89.8% branch coverage of the core packages, with CI running ruff, mypy --strict, pytest and a Docker check that the build toolchain is absent from the runtime image.

tweet-sentiment-analysis: NLP fine-tuning pipeline with Rust preprocessing

Python · PyTorch · HuggingFace Transformers · Rust · Polars · github.com/LukeSantossz/tweet-sentiment-analysis

- Delivered 0.887 macro F1 / 92.3% accuracy against a frozen-features linear-probe baseline at 0.584 / 67.5%, a 51.9% relative gain concentrated in the rarest classes, after per-class error analysis drove a pivot in task and backbone.
- Rewrote preprocessing as a Rust CLI benchmarked at 28.5x the Python reference on 1M rows (33.08s to 1.16s), with output parity validated at three scales.

Open source

- **my-framework** (MIT), github.com/LukeSantossz/my-framework: engineering-standards framework whose pre-push cross-provider review gate is a pluggable adapter chain telling "reviewed" from "reviewer unavailable". 101 shell tests; reused across five repositories.
- **AutoMQ**, github.com/AutoMQ/automq PR #3261: accepted fix in a Kafka-compatible streaming platform, adding response-size validation so a count mismatch fails pending requests cleanly instead of raising IndexOutOfBoundsException; 157 lines of tests for a 6-line fix.

SKILLS

Languages: Python, SQL, TypeScript, Rust

LLM, RAG & Agents: RAG, LangGraph, LangChain, agent orchestration, LLM-as-judge evaluation, semantic entropy, embeddings, Anthropic API, OpenAI API

ML: PyTorch, HuggingFace Transformers, scikit-learn, fine-tuning

Backend & Data: FastAPI, PostgreSQL, pgvector, Qdrant

Cloud & Practices: Docker, GCP (Cloud Run, BigQuery, Dataflow), GitHub Actions (CI/CD), pytest, Test-Driven Development

PUBLICATIONS & AWARDS

- **"Soil Texture Classification by Image: Deep Feature Learning vs. Handcrafted Methods for Precision Agriculture"**, poster, 17th ICPA / 11th ConBAP, Porto Alegre, July 2026.
- **3rd place, 16th FETEPS 2025** (Centro Paula Souza Technology Fair), among 1,300+ entries.
- **FAPESP grant 2024/00985-1**, volunteer researcher on the funded project behind SmartB100.

EDUCATION

Fatec Shunji Nishimura | Technology Degree (CST), Big Data in Agribusiness | 01/2024 - 11/2026

Languages: English (B2, Upper Intermediate) · Portuguese (Native)